

Transforming Certification for Better Care

ABMS CONFERENCE 2026

What AI Gets Right – And Wrong – About Literature Reviews

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Disclosure

The presenters have no conflicts of interest to disclose.

Introductions

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Session Overview

- Objectives
- Case Studies
 - 4 Case Studies
 - Examples of use AI in Literature Searches
 - Use of AI to Mitigate issues AI creates
- Discussion

Objectives

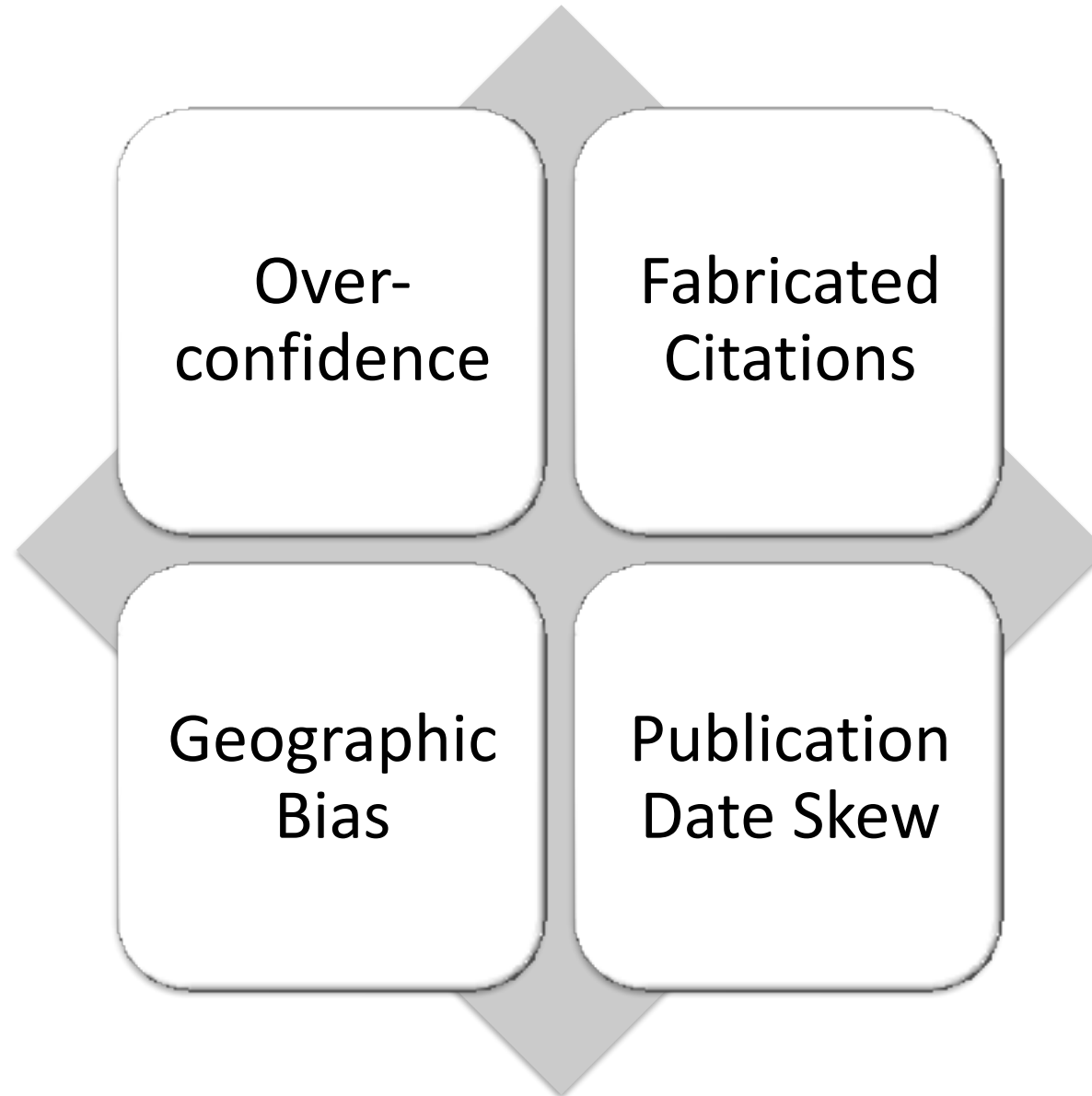


How does AI change the shape of literature reviews?



What does it mean for evidence-based decision-making?

Case Studies



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Case Study 1

AI and the Interpretation of Evidence:
Or, How Uncertainty Quietly Disappeared

Case Study 1 – Overconfidence



Scenario

A committee member receives an AI-generated summary of a study on [Endotracheal Intubation](#) just before an important discussion. It is concise, persuasive, and seems ready for decision-making. Only later does someone notice that the original researchers repeatedly warned their findings were preliminary. The summary got the facts right but overstated the confidence.



Do you use an AI tool to summarize articles for you?



Which one(s)?



Has your experience been positive? Negative?

Case Study 1 – Overconfidence

What Makes It Convincing

- Articulate and well-structured
- Clear, concise conclusions
- No obvious factual errors
- Easy to read and understand

What Deserves Scrutiny

- More confidence than the evidence supports
- Limitations and uncertainty are minimized
- Mixed findings may appear settled

Example of an AI Summary

Ryason and colleagues developed and evaluated a formative assessment tool for endotracheal intubation performance....

.....The study suggests that the tool provides a **valid and reliable way to assess endotracheal intubation competency** and deliver structured feedback to learners.....

.....**Overall, the findings support** the use of the assessment instrument for evaluating technical performance during endotracheal intubation.

Actual Article Conclusion

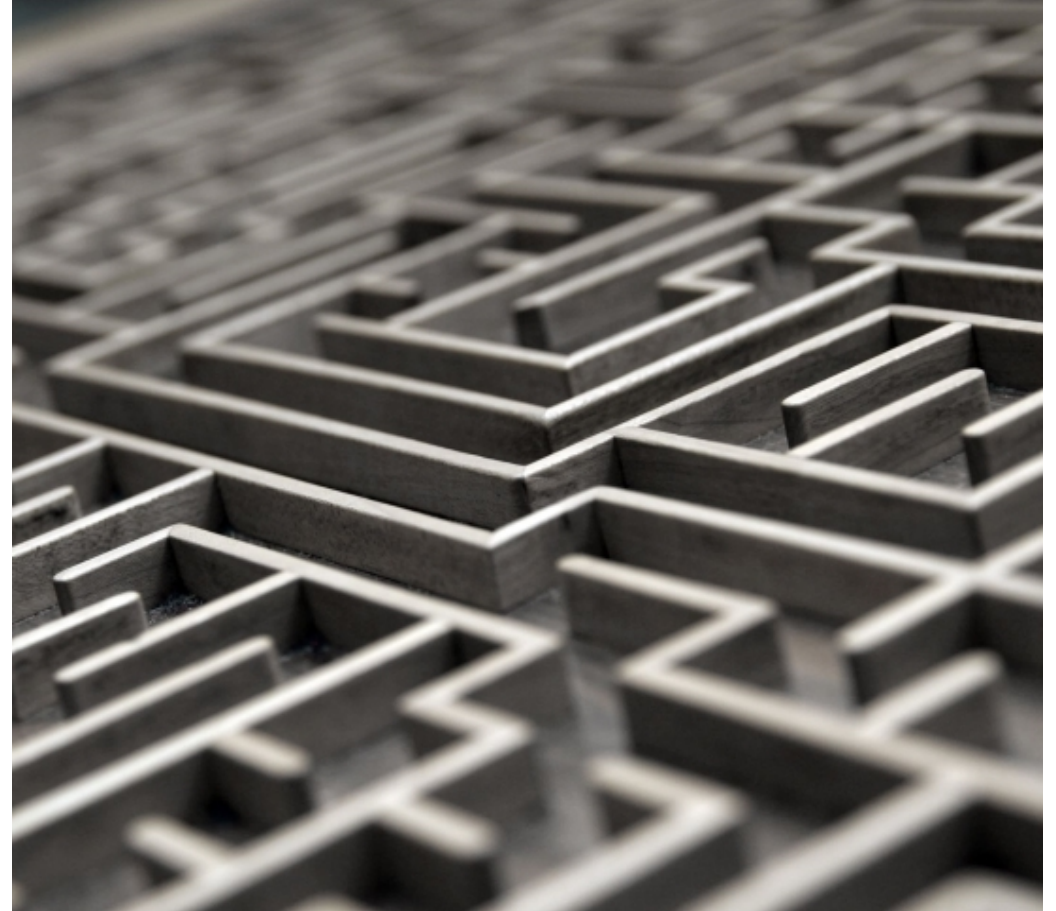
In conclusion, the results of these studies provide evidence that the checklist metric items and descriptions developed by our experts have **strong measurement** characteristics.....

.....Use of the metric **may** take place in training within simulators and the operating room to give feedback to individuals.....

.....**Further testing is needed** to demonstrate utility and effectiveness as a feedback tool. The metric **potentially provides** a way to give more detailed technical feedback than currently available ETI checklists.

Case Study 1 – Overconfidence

Mitigations –
Ask AI to
Identify the Nuances



Case Study 1 – Overconfidence

Mitigation Strategy #1: Ask more specific questions.

Instead of:

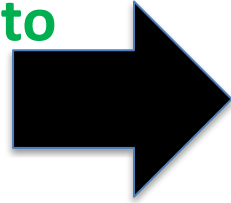
“Summarize the main findings?”

Ask:

“Summarize the findings *and explicitly describe the limitations, uncertainty, and confidence level of the evidence.*”

Summarize

Ryason and colleagues developed and evaluated a formative assessment tool for endotracheal intubation performance....The study suggests that the tool provides a **valid and reliable way to assess endotracheal intubation competency** and deliver structured feedback to learners. **Overall, the findings support** the use of the assessment instrument for evaluating technical performance during endotracheal intubation.



Summarize and...

Ryason and colleagues developed an assessment tool... The findings provide **promising evidence** that the tool can support structured assessment and feedback. However, **the study does not establish that this is the best method** for assessing competency, **nor does it demonstrate improvements** in clinical outcomes or patient safety. **Overall, confidence in the evidence is moderate, and additional research is needed** to compare this approach with alternative assessment method.

Case Study 1 - Overconfidence

Mitigation Strategy #2 - Try different AI tools.

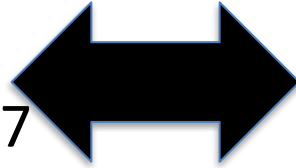
- Elicit
 - AI research assistant that searches academic literature and identifies relevant research studies.
 - Generates evidence summaries, literature reviews, and structured analyses based on published papers.
- NotebookLM (Google)
 - AI research assistant that analyzes source materials you upload or specify.
 - Generates summaries, study guides, Q&A, and insights grounded in those sources.

Elicit

“The evidence **supports** the metric as a promising tool for structured observation and feedback, but **not yet as a definitive** competency examination. The study used only 27 usable videos, expert raters judged **recordings rather than live** procedures, all procedures used a Mac 3 C-MAC blade, and the **authors had not established a validated score** separating competent from non-competent performance.”

NotebookLM

“By analyzing synchronized video recordings of actual clinical procedures, the study **confirmed** that the metric's scores align closely with **expert global rankings and ratings**. Statistical modeling further identified a reduced 3-item set that serves as a **highly accurate predictor** of overall clinical competency.”



Case Summary – Overconfidence

- When AI summarizes evidence, uncertainty and nuance can quietly disappear.
- Assessing the strength of evidence remains a fundamentally human responsibility.



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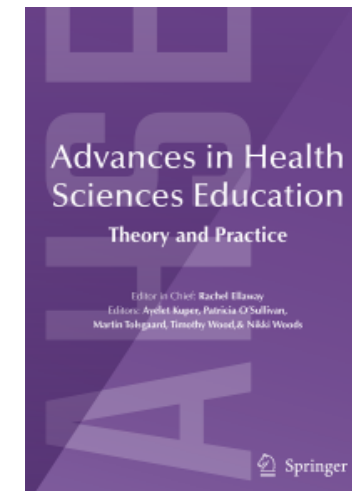
Case Study 2

AI-Assisted Citation Generation and Verification:
Or, The Citation That Never Existed

Case Study 2 - Fabricated Citations

Scenario

With a board report due the next morning, a colleague shares an AI-generated literature review complete with polished citations from reputable journals. The evidence appears well supported until you begin checking the references. Some articles cannot be found, while others contain incorrect authors, titles, or publication details.





Share stories of reference hallucinations at your table



How did you handle checking for reference validity?



What might you do differently after hearing others' experiences?

Case Study 2: Fabricated Citations

What Makes It Convincing

- Many valid and useful citations are identified.
- Some obscure previously unknown researchers are found.
- The author is well regarded.

What Deserves Scrutiny

- One of the citations contains hallucinated metadata
- One citation is used out of context.

Types of Hallucinations – “Easy” to Investigate

Completely Fabricated Citation

- The article, authors, or journal do not exist.

Real Journal, Fake Article

- The journal exists, but the cited article was never published.

Metadata Hallucination

- Real article, but incorrect year, volume, issue, pages, author list, or DOI.

Citation Mashup

- AI combines parts of multiple real citations into a single nonexistent reference.

Invented DOI

- DOI appears legitimate but does not resolve to a real publication.

Mismatched DOI

- DOI resolves to a real article, but not the article being cited.

Types of Hallucinations – More difficult to Confirm

Misinterpreted Findings

- The paper exists, but the AI gets the findings wrong.

Example:

Paper finding:

Residents reported that an AI tutor was easy to use and engaging. (Jones et al., 2026)

Misused Citation

- The paper exists, but it doesn't support the claim.

Misinterpreted findings:

The study demonstrated that AI tutoring improved examination scores.

Misused citation:

AI tutoring improves examination scores (Jones et al., 2026)

Case Summary – Fabricated Citations

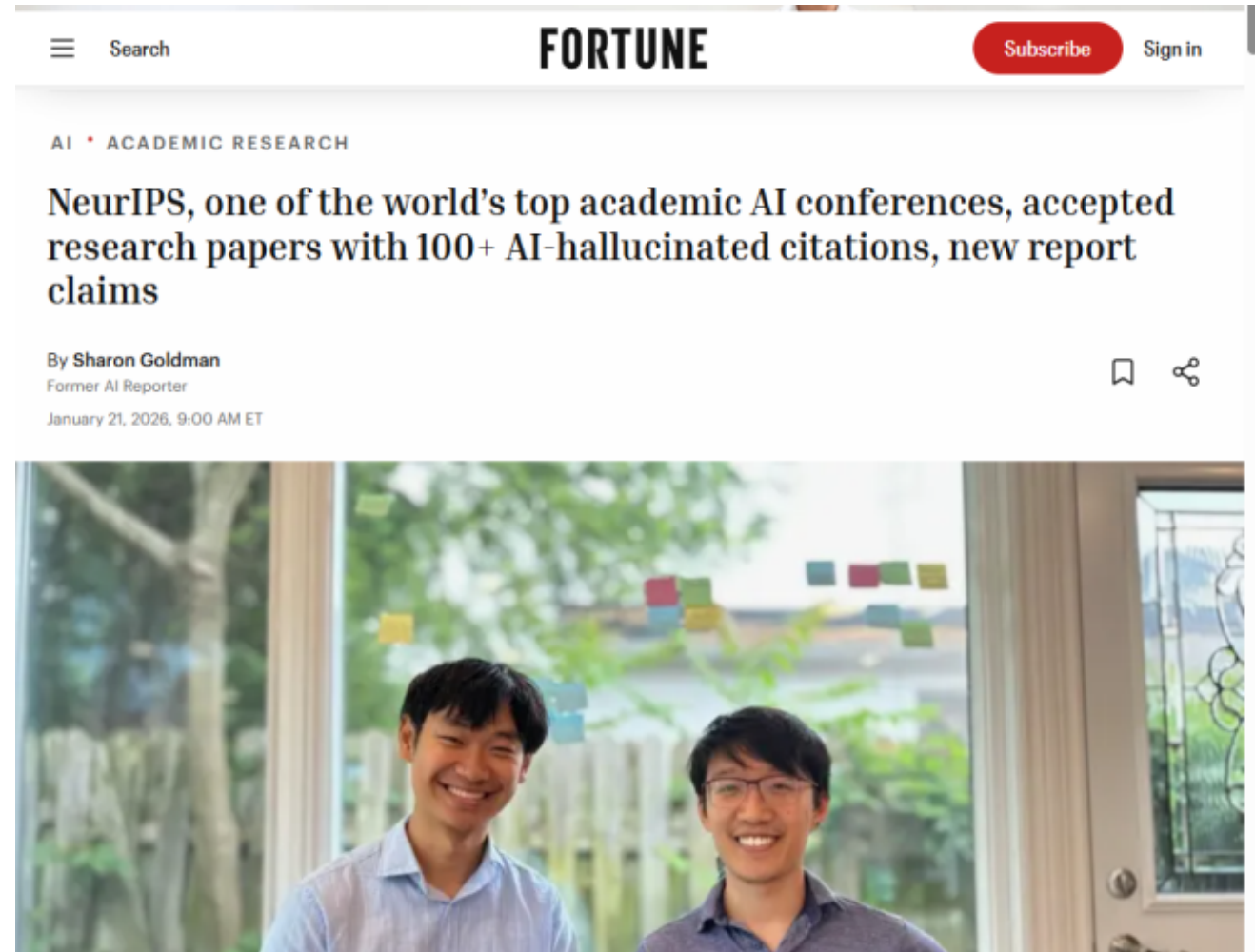
AI citations can be wrong for 3 reasons:

- The source doesn't exist
- The citation details are wrong
- The source exists but doesn't support the claim

Still Hallucinating?

- Top Academic AI conference accepts papers with hallucinations (January 2026)

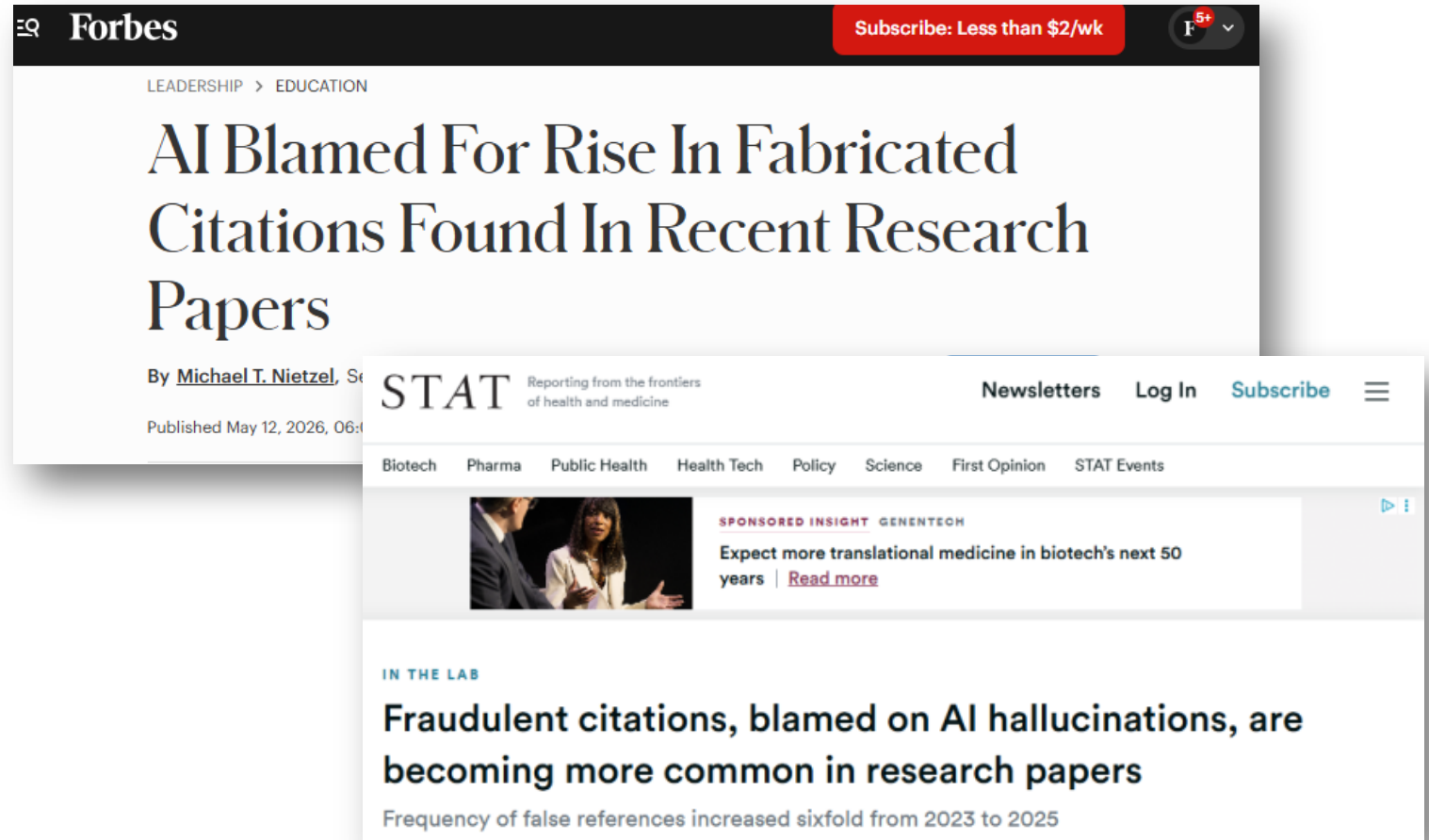
<https://fortune.com/2026/01/21/neurips-ai-conferences-research-papers-hallucinations/>



Still Hallucinating?

Forbes and STAT cited
“Fabricated citations: an
audit across 2.5 million
biomedical papers” (The
Lancet, May 2026)

In early 2026,
1 in 277 papers had at least
one fabricated reference



Still Hallucinating?

The Lancet,
May 2026
article also
showed that 57
per 10000
papers are
entirely
fabricated

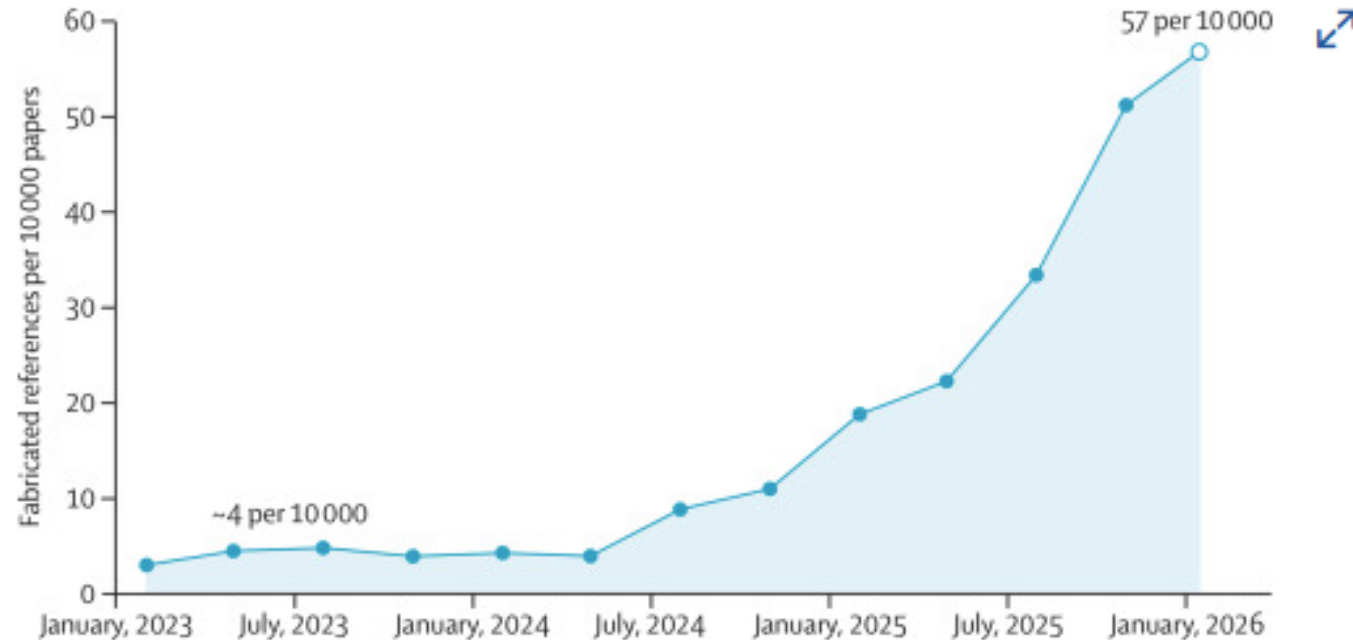


Figure Quarterly rate of fabricated references per 10 000 papers in PubMed Central from January, 2023, to February, 2026

[Show full caption](#) ✓

Case Study 2 – Fabricated Citations

Mitigation Strategy #1 –
Ask AI to
double-check citations



Case Study 2 – Mitigation Strategy #1

- Start from verified citations or seed papers
- Upload to AI
- Prompt:
“Verify the citations in this paper. For each reference, determine whether it exists, whether the bibliographic information is accurate, and whether the cited source supports the claim being made. Flag any fabricated, inaccurate, or potentially misleading citations and summarize the results in a table.”

Don't just ask AI to find citations. Ask AI to verify citations.

Recent use of AI to detect hallucinations



CiteCheck

✓ Checked against 150M+ scholarly records

Catch fake citations *before you submit.*

AI drafts invent references that look perfect and don't exist. Paste your reference list and CiteCheck verifies every entry against real academic libraries — then hands you a corrected list.

[Check your citations free →](#)

[How it works](#)

No account needed to try · 3 full-quality checks free · Anonymous checks aren't stored — sign in to save your history



Devlin, J., et al. (2019). BERT: Pre-training of deep bidirectional transformers... NAACL-...
Matched a real record in Crossref.



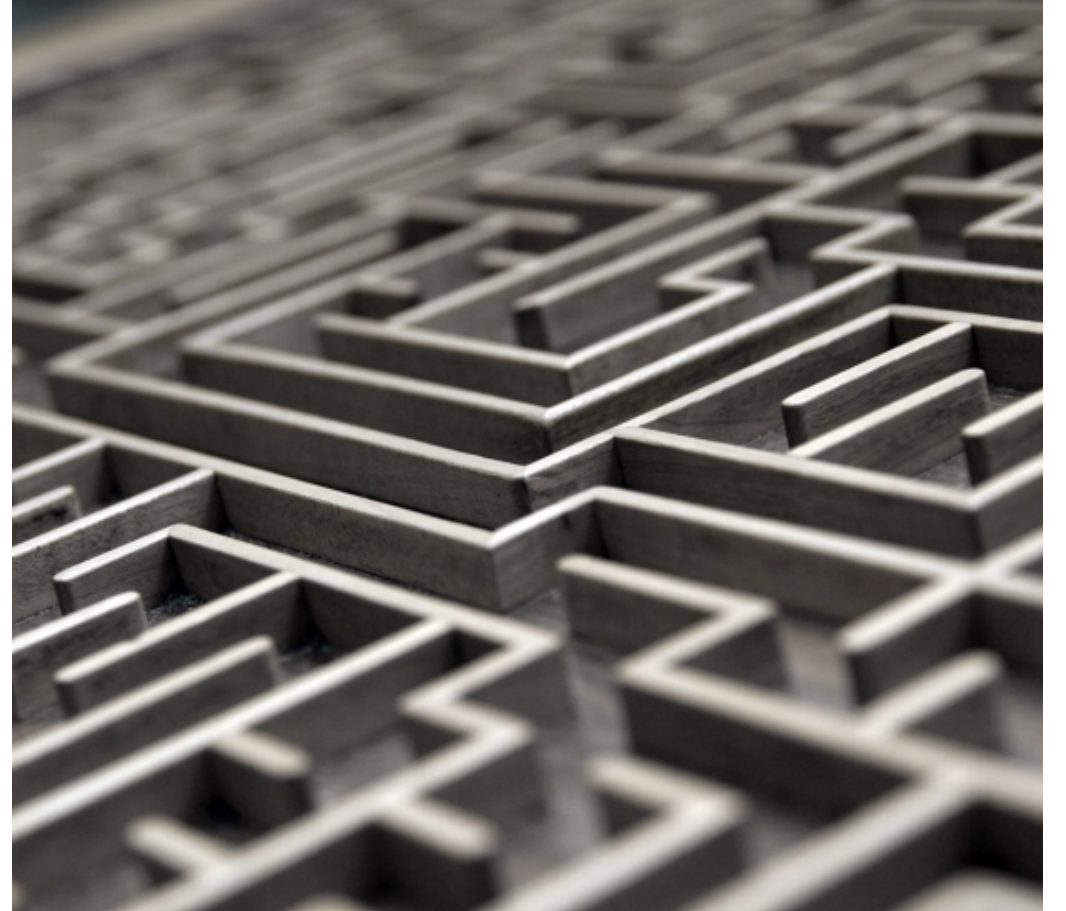
LeCun, Y., et al. (2018). Deep learning. Nature, 521(7553), 436–444.
Real paper — but the year should be 2015.



Thompson, R. K. (2021). Neural coherence in transformer-based citation synthesis. J. C...
No matching record found. Likely fabricated.

Case Study 2 – Fabricated Citations

Mitigation Strategy #2 –
Ask AI to
develop the initial search strategy



Case Study 2 – Mitigation Strategy #2

Ask Copilot:

- Act as an expert medical librarian.
- I am conducting a literature review on whether simulation-based training improves endotracheal intubation performance among medical learners.
- Create a PubMed search strategy using:
 - Keywords
 - Synonyms
 - MeSH terms where appropriate
 - Boolean operators (AND, OR)
- Limits suitable for a peer-reviewed literature review
- Provide the final search strategy in a format that can be copied directly into PubMed.

Sample partial result

```
(  
  "Endotracheal Intubation"[Mesh]  
  OR intubation  
  OR airway management  
  OR laryngoscopy  
)  
AND  
(  
  simulation training  
  OR simulation-based education  
  OR "Simulation Training"[Mesh]  
  OR mannequin training  
)  
AND...
```

Case Summary - Fabricated Citations



- Successful researchers have never relied on perfect tools. They rely on thoughtful evaluation.
- AI can help us find more evidence, explore more questions, and work more efficiently.
- Our responsibility is to ensure that speed never replaces judgment.

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Case Study 3

Assessing Coverage and Representation in AI-Generated Reviews: Or, How Completeness Became Assumed

Case Study 3 – Geographic Bias

Scenario

Facing a tight deadline, a team turns to AI to generate a comprehensive literature review on assessment methods. The resulting report is polished, concise, and full of citations, making it easy to trust. Only later does a subject matter expert discover that key studies from international perspectives are missing. The problem wasn't inaccurate information; it was the illusion of completeness.



Case Study 3: Geographic Bias

What Makes It Convincing

- The review appears thorough.
- The citations are legitimate.
- The results are well founded.
- Nothing signals bias (if you're not looking).

What Deserves Scrutiny

- Studies favor a narrow geographic region
- Review excludes non-English evidence

Geographic Bias in AI

Machine Learning (2025) 114:263
<https://doi.org/10.1007/s10994-025-06916-9>

Evaluation of geographical distortions in language models

Rémy Decoupes^{1,2} · Roberto Interdonato^{1,3} · Mathieu Roche^{1,3} ·
Maguelonne Teisseire^{1,2} · Sarah Valentin^{1,3}

- The study evaluated **10 models** using four indicators
- Models showed a consistent **Western geographic bias**

Geographic Bias is not a new problem...

Journal of Librarianship and Information Science

Impact Factor: 2.4
5-Year Impact Factor: 2.6

[Journal Menu](#)

 Restricted access | Research article | First published online February 19, 2025

[Re-use permissions](#) 

Continued struggle: The growth of Scopus-indexed journals in Africa over the last decade

[Martin Elihaki Kanyika](#)   and [Eungi Kim](#) [View all authors and affiliations](#)

[Volume 58, Issue 1](#) | <https://doi.org/10.1177/09610006251319546> | [View article versions](#)

Case Study 3 – Geographic Bias

Mitigation Strategies –
Ask AI to
look for the gaps



Case Study 3 – Mitigation Strategy #1

Ask AI for a Geographic Citation Audit

AI Prompt: Review the reference list and identify potential blind spots. Are there geographic regions, disciplines, research traditions, or influential authors that appear to be missing?

Sample AI Response:

Potential geographic and language **bias detected**. The review appears to rely primarily on English-language publications and searches only PubMed. Most cited authors are based in North America, with limited representation from international research groups. **The review also does not discuss whether its findings apply across different healthcare systems, cultures, or geographic regions.**

Case Study 3 – Mitigation Strategy #2

Ask AI Who the Missing Voices Are

AI Prompt: Which authors, institutions, countries, or research groups are commonly cited in the assessment literature but do not appear in this review?

Sample AI Response:

The review appears to be dominated by North American and UK sources. Influential assessment research from Dutch, Canadian, Australian, and Asian groups is largely absent, including several commonly cited authors in programmatic assessment and CBME. **This pattern may suggest geographic bias in the evidence synthesis.**

Case Summary – Geographic Bias

- AI did not create geographic bias in the literature.
- Diverse evidence rarely emerges by accident.
- Used thoughtfully, AI can help surface missing voices and viewpoints.

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Case Study 4

Defining 'Recent' in AI-Generated Evidence
Summaries: Or, How 'Recent' Stopped Meaning
Recent

Case Study 4 – Publication Date Skew



Scenario

With a report due tomorrow, you find a literature review on PubMed that was published just last month. It looks current, authoritative, and exactly what you need. After citing its conclusions, you discover that the review relied heavily on AI-generated synthesis and missed several newer studies that may have changed the story. The publication date was fresh. The evidence was less so.

Case Study 4 – Publication Date Skew

What Makes It Convincing

- Review was published recently
- Found on PubMed
- Comprehensive reference list
- Written by recognized experts

What Deserves Scrutiny

- Missing the most recent studies
- Limited discussion of contradictory evidence
- Completeness of the evidence searched

Case Study 4 – Publication Date Skew

Mitigation Strategy –
Ask AI to
look for what is missing



Case Study 4 – Mitigation Strategy

Example Prompt:

Before I rely on this literature review, help me verify its completeness and currency.

- Analyze the publication years of the cited references.
- Identify any recent, highly cited publications on this topic that are not included.
- Determine whether important evidence may have been published after the review's search period.
- Highlight any gaps that could change the overall conclusions.
- Summarize the findings in plain language.

Case Study 4 – Mitigation Strategy

Sample AI Response:

Publication Date Analysis

- Most cited studies were published between **2013-2020**.
- Only **2 of 45 references** were published within the last two years.
- The evidence base may not fully reflect recent developments in the field.

Potential Impact on Conclusions

- Newer evidence challenges some assumptions presented in the review.
- Inclusion of recent studies could strengthen, modify, or qualify the review's conclusions.

Case Summary – Publication Date Skew

- The review was new. The evidence was not.
- Missing recent studies can alter conclusions and recommendations.
- Used thoughtfully, AI can help identify when a review may no longer reflect the current evidence.

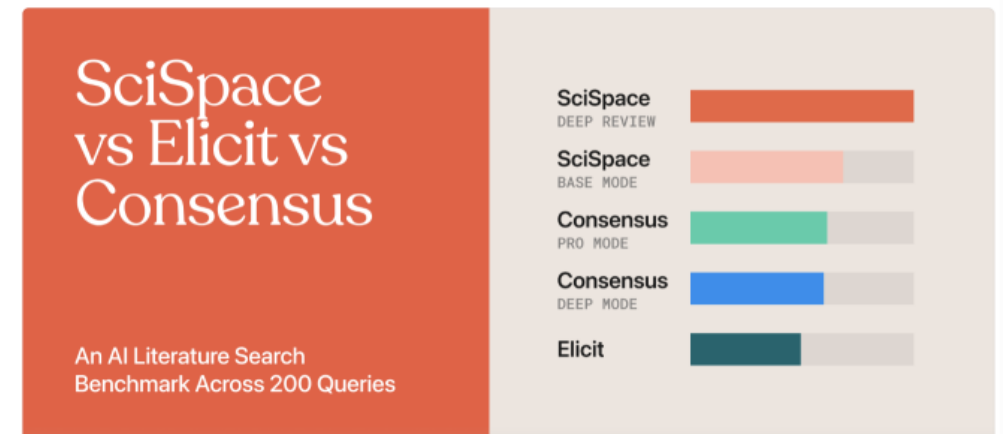
Current State of AI Literature Search

whitepapers

SciSpace vs Elicit vs Consensus: An AI Literature Search Benchmark Across 200 Queries

We benchmarked SciSpace, Elicit, and Consensus on 200 complex research queries, scored by three independent AI judges. The result: SciSpace Deep Review returned the most highly relevant papers — 26.3 per query on average, versus 13.0 for Elicit.

Jun 8, 2026 — Prudhvi



A note on transparency: This benchmark was run by the SciSpace research team. To keep scoring as neutral as possible, relevance was judged by three external frontier models rather than by SciSpace, and the full methodology is documented below so you can evaluate it for yourself.

Final thoughts...

Literature reviews are still about finding and evaluating information.

What's changed is...

- Larger scale/total number of papers
- Unit of analysis/allows for analysis of combined units of papers
- Searching is conversational/no need for strict search terms
- Easier to analyze beyond just studies/authors, institutions, citations
- An abundance of information
- Personalized review reports



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